

Abhay Kumar

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Nationality: Indian

Summary

Master's Physics student specialising in Materials Science and Electronics with a strong foundation in physics and a keen interest in translating scientific research into practical technological innovations. Aspiring to bridge the gap between core physics and engineering by developing advanced materials and electronic devices for future applications. Particularly interested in sustainable energy technologies, functional materials, semiconductor devices, and energy storage systems.

Education

🎓 M.Sc. in Physics	2024 – 2026
🏛 J.C. Bose University of Science and Technology, YMCA, Faridabad	
Relevant Coursework: Materials Science, Electronics, Quantum Mechanics	
🎓 B.Sc. (Hons.) in Physics	2021 – 2024
🏛 J.C. Bose University of Science and Technology, YMCA, Faridabad	
Relevant Coursework: Mathematical Physics, Atomic Physics, C++, HTML	

Experience & Technical Training

Workshop on Arduino and Embedded Systems May 2026
2-Day Hands-on Training in Wave Techtronics
Explored microcontroller architectures, sensor integration, and circuit automation basics relevant to building prototype electronic devices.

Workshop on Fabrication & Characterisation of Electronic Materials and Devices April, 2023
5-Day Intensive Technical Training – Advisor: Dr Anuj Arya (Associate Professor), JC BOSE UST
Gained practical insights into the synthesis, fabrication techniques, and advanced characterisation tools used for modern electronic materials and semiconductor devices.

9th Indian International Science Festival (IISF 2023) Dec 2023
4-Day Participant
Engaged with global scientific discussions, attended mega-science exhibitions, and explored emerging frontiers in science and technology adaptation.

Research & Projects

Fabrication & Characterisation of Bio-Based Piezoelectric Thin Films Using Aloe Vera

M.Sc. Dissertation — Advisor: Prof. Anuradha Sharma (Department of Physics)

- Engineered eco-friendly, flexible piezoelectric thin films using extracted active components from Aloe vera for self-powered biocompatible electronics.
- Optimised gel recovery using high-speed centrifugation (6,000 rpm at 10°C) and leveraged ultrasonic-assisted extraction to enhance the diffusion kinetics of aloin.

- Investigated room-temperature piezoelectricity driven by the long-range molecular ordering and hydrogen bonding of acemannan chains.
- Deployed processed dry films in energy harvesters to convert biomechanical energy into electricity, achieving a peak-to-peak voltage generation of 600 mV (± 300 mV).

Awards & Positions of Responsibility

Student Coordinator

2025-2026

Department of Physics, J.C. Bose University of Science and Technology

Managed departmental events, facilitated smooth communication between faculty and students, and led student teams for academic seminars.

Active Volunteer

2021 – 2024

National Service Scheme (NSS) & Community Initiatives

- Nanhi Udaan NGO: Contributed to the Annapurna Project by providing free education and distributing nutritious meals to underprivileged children.
- Save Aravali NGO: Engaged in local environmental conservation drives, tree plantation initiatives, and sustainability awareness campaigns.
- Community Health: Recognised for supporting emergency healthcare needs through voluntary blood donation drives.

Skills & Languages

Laboratory Expertise:

UV-Vis Spectroscopy, FTIR Spectroscopy, Centrifuge Machine, Ultrasonic Bath, Muffle Furnace, Hot Air Oven, Magnetic Stirrer, pH Meter, Digital Weighing Balance and basic Physics laboratory devices.

Programming & Software:

C, HTML, CSS, Good command of AI tools.

Languages Spoken:

English (Fluent), Hindi (Native).